1 Scope

The question is what the sweep of 16 October found at the top venues for multi-party learning in 2023 to 2026, and how its papers state their adversaries. We count the search from the screening record, the coded papers by venue, year, kind, setting, protection and evaluation, and the ledger's coalitions by seven templates of role, choices and goal. We leave HE-OFT out, because the study covers the field.

2 Result

The sweep coded 349 papers, and the typical one studies horizontal federated learning on MNIST or CIFAR-10 against one of seven adversary templates (Tables 3, 4 and 6). 192 papers train by horizontal federated learning or federated fine-tuning, 215 evaluate on MNIST or CIFAR-10, and the templates hold 513 of 516 coalitions. The privacy line and the robustness line rarely meet in one threat model. 103 of 145 rows of clients that steer through their updates name no confidentiality mechanism, and 12 of 102 rows of a reading server face a check. The coder had to infer static or adaptive corruption in 374 of 516 rows, and 17 rows have an adaptive adversary (Table 8). Per paper, the HE share rose from 11 to 19 per cent from 2023 to 2026 (Fisher p of 0.241), and the deviating share held near 55 per cent (Table 5).

Does not show: Every count is empirical, on rows that one language-model coder extracted from each paper, and these rows await a check against their papers. The candidates are only the papers a screener recorded, and screening recall is therefore unmeasured. We did not screen EuroS&P, SaTML or CSF. The 2026 counts lack NeurIPS 2026 and may lack part of CCS 2026. Templates, word families, mechanisms and checks are keyword matches on the coded cells, by rules the script states.

3 How it was produced

Sources	`lib/literature/screening.csv` at commit 5e8872b, and `lib/literature/ledger.csv` as of 2026-10-16 12:00, which differs from that commit only in the note paths
Range	2023 to 2026 at IEEE S&P, USENIX Security, CCS, NDSS, PoPETs, CRYPTO, EUROCRYPT, ASIACRYPT, NeurIPS, ICML and ICLR, with three anchors from 2021 and 2022
Screened	573 candidates, 349 papers coded on 516 ledger rows, HE-OFT left out
Artifacts	one CSV per table and a facts CSV in `lib/reports/src/data/`, under this report's stem
Reproduce	the tables script in `lib/reports/src/figures/`, run from the repository root

4 What this changes

This report supersedes the same day's sweep report, whose rising deviating share counts 70 rows with choices such as "none (semi-honest)" as deviating. Frame 9 of the meeting deck tests the HE trend on rows, which share their paper, and its p below 0.001 becomes 0.241 per paper under this report's rule. To study the field, take one template at a time, read its anchors in Table 9, then filter the ledger to its rows. Four things the field seldom states belong in any threat model we write: static or adaptive corruption, an honest or dishonest majority, when the adversary commits its choices, and what it knows of honest updates.

5 Method

The search

Six language-model screeners read the accepted-paper lists of the eleven venues for 2023 to 2026, one screener per venue group. A screener kept a paper when two or more parties train or fine-tune

Table 1. The search per venue. Recorded counts the candidates, the papers a screener recorded as near the rule, and accepted papers dropped at a glance are not among them. Dropped counts candidates dropped at screening, Read out those dropped after reading, and No PDF those kept without an accessible PDF. Coded papers have a PDF, a note and ledger rows, and the year columns split them. Earlier holds three anchors from 2021 and 2022.

Venue	Recorded	Dropped	Read out	No PDF	Coded	2023	2024	2025	2026	Earlier
IEEE S&P	39	13	0	1	25	9	7	3	5	1
USENIX Security	61	20	2	0	39	10	11	12	6	0
CCS	78	34	1	2	41	5	14	14	6	2
NDSS	39	16	0	0	23	3	5	9	6	0
PoPETs	56	15	0	0	41	12	6	14	9	0
CRYPTO	5	2	0	0	3	0	1	2	0	0
EUROCRYPT	3	2	1	0	0	0	0	0	0	0
ASIACRYPT	13	6	1	1	5	3	0	1	1	0
NeurIPS	87	36	2	0	49	17	15	17	0	0
ICML	119	44	6	0	69	18	21	18	12	0
ICLR	73	18	1	0	54	12	15	10	17	0
all eleven	573	206	14	4	349	89	95	100	62	3

Table 2. Why screeners dropped candidates, matched by keyword on the reason each screener wrote. The first matching reason counts.

Reason given at screening	Papers
one party trains, or one trained model is attacked	76
no adversary, or privacy and security only as motivation	49
no model is trained	40
inference or serving only	25
cryptography not tied to learning	2
outside the main track	1
other reasons	13

on data or models they hold, and the contribution concerns privacy or security against an adversary. Each screener recorded the papers near the rule with a reason, and the screening record holds 573 such candidates (Table 1). Accepted papers far from the rule left no record. Screeners dropped 206 candidates, most often because one party trains or because the paper states no adversary (Table 2).

We placed each kept paper with its PDF and a note, after matching its title and authors on the first page. A coder then read each placed paper and wrote one ledger row per coalition on 22 columns. It dropped 14 papers as outside the rule, and 4 kept papers wait for an accessible PDF. A row records what a coalition sees, knows and sets, its goal, the protection, the checks it faces, the evidence, and any step between two adversaries the paper settles.

What the coded papers are

Defences and protocols outnumber attacks, and DP is the most common protection (Table 3). 146 papers name no protection for any coalition, and 55 of them are attacks and 69 defences, 41 of these by robust aggregation. Clients alone are the most common coalition, and the three goals appear in similar numbers (Table 4). The evaluations are small. 62 papers start from a public pretrained model, 57 reach a transformer as their largest model, and 16 run no experiment.

Trends by year

Counted per paper, no share moves enough to call a trend (Table 5). The HE share rose from 11 to 19 per cent from 2023 to 2026, with Fisher p of 0.241. The share with a deviating coalition went from 58 to 55 per cent, with p of 0.74.

Table 3. The coded papers by kind of contribution, setting and protection. A paper counts under each mechanism that any of its coalitions faces, so the protection column sums past the total.

Kind	Papers	Setting	Papers	Protection	Papers
defence	124	horizontal FL	172	none named	146
protocol	109	secure aggregation	49	DP	109
attack	74	MPC training	37	MPC	56
two kinds or more	17	vertical or split learning	29	secure aggregation	56
theory	15	decentralized learning	24	HE	48
SoK	6	federated fine-tuning	20	ZK	22
measurement	4	multiparty or threshold HE	14	TEE	7
		other	4		

Table 4. The coded papers by coalition, goal and evaluation. A paper counts under every coalition and goal its rows hold, and under every mark of its evaluation.

Coalition	Papers	Goal	Papers	Evaluation uses	Papers
clients alone	205	distinguish	151	MNIST or CIFAR-10	215
the server alone	124	invert	139	a public pretrained model	62
the server with clients	68	steer	132	a transformer	57
a querier or eavesdropper	34	other	28	token sequences	26
				no experiment	16

Table 5. Per cent of each year's coded papers. Deviating counts papers with a coalition that sets anything beyond queries. A mechanism column counts papers that name it for some coalition. Proven counts papers with a proven or sketched result. The 2026 column lacks NeurIPS 2026.

Year	Papers	Deviating	HE	DP	SecAgg	MPC	ZK	TEE	Proven
2023	89	58	11	30	18	16	8	1	73
2024	95	61	11	33	11	17	2	1	77
2025	100	55	16	37	17	20	8	3	82
2026	62	55	19	23	19	10	8	3	77

The seven templates

A row's template follows from its role, its choices and its goal. The server alone splits by its choices, and clients alone split by goal and then by choices. The server with clients forms one template, and a querier or an eavesdropper outside training forms another. Three rows name a trainer, a platform operator or no role, and they stay outside.

The server that reads what it receives follows the protocol and inverts or distinguishes from updates, aggregates and models. DP is its usual protection and its usual proof (Table 7). The papers give this server the architecture in 64 of 93 inverting server rows, and surrogate data in 18. The two gradient-leakage SoKs grade this server by levels of knowledge that are not nested.

The server that sets what clients compute sends a crafted model, a changed architecture, or a different model to each client. An update or a sum of updates then reveals a client's data. Prose and constructive lower bounds are its most common forms of evidence. 24 of its 48 rows require or measure stealth, the highest share of any template.

The server with clients is the template of cryptographic protocols, namely secure aggregation, MPC and multiparty HE. 74 of its 78 rows name a mechanism, and 37 face a check. Its papers prove standalone simulation security most often. They state static or adaptive corruption in 27 rows, the highest share of any template.

Clients that set their updates form the largest template, in the papers' words Byzantine or malicious clients. The paper assumes an honest majority in 107 of 145 rows, and the usual result is a convergence bound for a robust aggregation rule. These rows evaluate an adaptive adversary

Table 6. The seven templates in ledger rows. Choices and goals give the two most common values, and a row may hold several. Message is a node output such as an update or a decryption share. Bare counts rows that name no protection mechanism. Check counts rows facing an input check, a proof of honest computation, a robust aggregation rule or a detector. Proven counts rows with a proven or sketched result. Adaptive counts rows evaluating an adversary that adapts to the defence, out of the rows that say.

Template	Rows	Papers	Choices	Goals	Bare	Check	Proven	Adaptive
server that reads what it receives	102	91	nothing 99, data 3	invert 58, distinguish 55	28	12	77	13 of 42
server that sets what clients compute	48	40	model 29, message 16	invert 35, distinguish 9	13	21	32	7 of 20
server with clients	78	68	nothing 42, message 35	distinguish 55, invert 36	4	37	75	4 of 21
clients that set their updates	145	127	message 133, data 85	steer 126, other 20	103	84	93	66 of 120
clients that set only their data	21	21	data 21	steer 19, other 2	16	10	9	8 of 19
clients that read the model	85	72	nothing 62, message 14	distinguish 52, invert 42	22	14	67	7 of 31
querier or eavesdropper	34	34	nothing 30, data 3	distinguish 24, invert 10	8	7	28	4 of 19

Table 7. What each template assumes and how it supports its claim, in rows. Evidence gives the two most common forms of the result. Honest and dishonest count rows assuming an honest or a dishonest majority. Stated counts rows in which the paper itself says static or adaptive corruption. Stealth counts rows that require or measure a stealth constraint.

Template	Evidence	Honest	Dishonest	Stated	Stealth
server that reads what it receives	DP 33, standalone simulation 9	17	6	5	4
server that sets what clients compute	prose 7, constructive lower bound 7	7	6	4	24
server with clients	standalone simulation 31, DP 12	28	26	27	2
clients that set their updates	convergence 33, empirical 21	107	18	17	68
clients that set only their data	empirical 9, generalization by stability 1	16	1	2	9
clients that read the model	DP 25, standalone simulation 22	33	20	18	8
querier or eavesdropper	DP 17, game 5	3	1	3	2

more often than any other template. Clients that set only their data poison it before training and follow the protocol after, and their evidence is mostly empirical.

Clients that read the model learn about another party's data from what they receive. The template mixes parties of MPC training, which prove simulation security, nodes of decentralized learning, which prove DP, and parties of vertical learning. A querier or an eavesdropper observes the trained model or the messages from outside, most often to distinguish, and DP is again its usual protection.

What a word means

The paper's word for its adversary fixes the choices only together with the role. A malicious server sends a crafted model in 24 of 38 rows. Malicious clients set their data in 54 of 75 rows and their updates in 55, and Byzantine clients set their updates in 53 of 56. The ledger therefore stores what a coalition sees, knows and sets, and keeps the word as provenance.

Table 8. How each community states its threat models, in per cent of its rows. Formal counts statements as an ideal functionality or a game, Theorem as the assumptions of a theorem, and Prose as prose or a taxonomy. The last four columns count rows in which the coder inferred a value the paper did not state. They are static or adaptive corruption, an honest or dishonest majority, what the coalition sets or when, and what it knows. Cryptography holds 17 rows.

Community	Rows	Papers	Formal	Theorem	Prose	Corruption	Majority	Choices	Knowledge
security	266	169	29	30	41	73	52	60	31
cryptography	17	8	94	0	6	35	47	65	47
learning	233	172	5	62	32	75	44	51	48

Table 9. Papers to read first in each template, by citation key. The choice is ours and favours the 33 papers marked for a full read.

Template	Read first
server that reads what it receives	DHWSGRC25 and CFMPV25, the two gradient-leakage SoKs (USENIX Security 2025). PDBMV24, DAGER (NeurIPS 2024)
server that sets what clients compute	PFA22, model inconsistency against secure aggregation (CCS 2022). ZSEEAB24, LOKI (S&P 2024). CFMPV26, detectability of active inversion (S&P 2026)
server with clients	BGLLMRY23, ACORN (USENIX Security 2023). JZDK25, AlphaFL (PoPETs 2025). REGB26, the integrity SoK (PoPETs 2026)
clients that set their updates	RT25, robustness against learning (S&P 2025). CKS24, attacking robust aggregation (S&P 2024). AGGPS23, the trilemma (ICML 2023)
clients that set only their data	SHKR22, poisoning in production (S&P 2022). BLGB26, stability bounds (ICML 2026)
clients that read the model	MCB24, attacks in decentralized learning (ICML 2024). YLGHP25, URVFL (NDSS 2025). RDGSM26, selective weight tampering (PoPETs 2026)
querier or eavesdropper	LLR23, passive membership inference in FL (ICLR 2023). ZLSGWYL26, membership defence (NDSS 2026)

What the papers leave unstated

Each community states its threat model in its own form (Table 8). Cryptography papers use an ideal functionality or a game, learning papers the assumptions of a theorem, and security papers prose as often as anything else. The coder inferred the corruption model in 374 rows and the majority in 249. Cryptography rows are the exception for corruption. Of the 164 steering rows, 76 state no stealth constraint, and 55 say whether the adversary sees honest updates. 69 of the 135 steering rows that report it evaluate an adaptive adversary. 114 rows record a step between two adversaries that the paper settles by a collapse or a separation.

What to read first

Table 9 names two or three papers per template. Each has a note, but every note except HE-OFT's still holds only the template text, and the ledger rows are the only extraction until a full read fills the note.